DJANGO 6.1 • FIELD OPERATIONS RUNBOOK

Project Manager: Inventory CRUD Walkthrough

Architecture, Route Registry, Model Schema, and Operational Workflows

HOST: **localhost:8000**
WORKSPACE: **Project_manager**
THEME: **Cadmium Green (#137547)**

1 COMPLETE CRUD ENDPOINT & CONTROLLER MATRIX

26 ACTIVE ROUTES

DOMAIN / MODEL	URL ROUTE & PATTERN	CLASS-BASED VIEW	TEMPLATE RESOURCE	ACTION / PERM
Dashboard	`/`	`views.home`	`templates/home.html`	KPI OVERVIEW
Part	`/parts/`	`PartListView`	`core/part_list.html`	SEARCH & FILTER
	`/parts/add/`	`PartCreateView`	`core/part_form.html`	CREATE (CKEDITOR)
	`/parts/<pk>/`	`PartDetailView`	`core/part_detail.html`	STOCK SPECS
	`/parts/<pk>/edit/` | `/delete/`	`PartUpdateView` | `DeleteView`	`part_form.html` | `confirm_delete.html`	UPDATE / PURGE
Company	`/companies/` | `/add/`	`CompanyListView` | `CreateView`	`company_list.html` | `form.html`	SUPPLIER INDEX
	`/companies/<pk>/` | `/edit/`	`CompanyDetailView` | `UpdateView`	`company_detail.html`	PARTS CASCADE
Manufacturer	`/manufacturers/` | `/add/`	`ManufacturerListView` | `CreateView`	`manufacturer_list.html`	OEM INDEX
	`/manufacturers/<pk>/` | `/edit/`	`ManufacturerDetailView` | `UpdateView`	`manufacturer_detail.html`	CATALOG VIEW
Category	`/categories/` | `/add/`	`CategoryListView` | `CreateView`	`category_list.html`	CLASSIFICATION
	`/categories/<pk>/` | `/edit/`	`CategoryDetailView` | `UpdateView`	`category_detail.html`	HIERARCHY TREE
Location	`/locations/` | `/add/`	`LocationListView` | `CreateView`	`location_list.html`	BIN LOCATIONS
	`/locations/<pk>/` | `/edit/`	`LocationDetailView` | `UpdateView`	`location_detail.html`	STOCK BY BIN

2 PART MODEL SPECIFICATION

CORE/MODELS.PY

- **Identification**: `part_number` (unique, auto-generated fallback), `supplier_part_number`, `description`.
- **Calculated Pricing**: `price_per` × `quantity` → auto-computes `price` in model `save()`.
- **Superseded Architecture**: Self-referential FK `superseded_by = models.ForeignKey('self', null=True, blank=True, related_name='supersedes')`. Allows seamless forward & backward lookup.
- **Stock Metrics**: `quantity_on_hand`, `quantity_on_order`, `quantity_committed`, `quantity_availible`, `quantity` (defaults to 0, `blank=True`).
- **Rich Notes**: Rendered using `CKEditor5Field('Notes', config_name='extends')`.

> **Pricing Hook:** When `price_per` and `quantity` are provided, `save()` computes `self.price = Decimal(self.price_per) * Decimal(self.quantity)`.

3 SUPPLIERS & TAXONOMY

RELATIONSHIPS

- **Company (Supplier)**: One-to-Many relation to parts (`company.parts.all()`). Stores website URL, account number, contact rep, phone, email, and procurement notes.
- **Manufacturer (OEM)**: One-to-Many relation to parts (`manufacturer.parts.all()`). Distinguishes vendor distributor from original maker.
- **Category**: Self-nested hierarchy via `parent = ForeignKey('self')` for deep breadcrumb trees (e.g. *Passives → Resistors → SMD 0805*).
- **Location**: Warehouse, shelf, rack, or bin code (`BIN-xxx`) tracking exact physical storage slots.

RELATED MODEL	FIELD ON PART	ON DELETE BEHAVIOR
`Company`	`part.company`	`SET_NULL` (prevents data loss)
`Manufacturer`	`part.manufacturer`	`SET_NULL`
`Category`	`part.category`	`SET_NULL`
`Location`	`part.location`	`SET_NULL`

4 DESIGN SYSTEM & THEME

CADMIUM GREEN

- **Base Palette**: Integrated `bootstrapcolors-custom.css` featuring Cadmium Green (`#137547`), Forest (`#054a29`), and Mint accents.
- **Responsive Sidebar**: Permanent 260px desktop drawer; responsive collapse into touch-friendly mobile offcanvas drawer.
- **Theme Switcher**: Dual Light/Dark mode via Bootstrap 5.3 `data-bs-theme` with persistent `localStorage` sync.
- **Interactive Forms**: Grouped card layouts, floating labels, select widgets with `-- Select --` prompts, and real-time JS price calculators.

5 FORMS & VALIDATION SAFEGUARDS

CORE/FORMS.PY

- **Self-Reference Loop Prevention**: In `PartForm` and `CategoryForm`, `__init__()` dynamically excludes `pk=self.instance.pk` from choices.
- **Quantity Sanitization**: Form cleaning hooks default empty inputs to `0`, eliminating validation rejections on stock counters.
- **WYSIWYG Asset Inclusion**: Form headers inject `{{ form.media }}` to load CKEditor 5 assets dynamically.
- **Safe Deletion**: Reusable `confirm_delete.html` displays item details and warning before executing `POST` purges.

6 OPERATOR WALKTHROUGH & TERMINAL RUNBOOK

STANDARD PROCEDURE

```
# 1. Activate environment and boot the local development server source .venv/bin/activate python manage.py runserver
127.0.0.1:8000 # 2. Add reference suppliers, manufacturers, and storage locations curl -X POST
http://127.0.0.1:8000/companies/add/ -d "name=DigiKey&website=https://digikey.com&contact_name=Support" curl -X POST
http://127.0.0.1:8000/locations/add/ -d "name=Bin A12&code=BIN-A12&description=Main Component Rack" # 3. Create a component
with automatic price multiplication curl -X POST http://127.0.0.1:8000/parts/add/ -d
"part_number=LM317T&quantity=100&price_per=0.75&confirmed=true" # -> Automatically calculates: price = 75.00 | redirects to
/parts/ # 4. Filter and query the parts database curl "http://127.0.0.1:8000/parts/?q=LM317&confirmed=true" # 5. Supersede a
legacy part with an updated replacement # In /parts/<legacy_pk>/edit/, select the new component in the 'Superseded By'
dropdown.
```

WORKFLOW PHASE	ACTION PERFORMED IN UI	SYSTEM REACTION & STATE CHANGE
1. Supplier Onboarding	Navigate to `/companies/add/`, fill in rep & contact details.	Creates `Company` record. Instantly selectable in the Part creation dropdown.
2. Part Registration	Go to `/parts/add/`. Enter part number, unit price, quantity, and specs.	Calculates total price on save. Shows success alert on `/parts/`.
3. Inventory Audit	Open `/parts/<pk>/` detail view.	Renders stock badge tiles (On Hand, On Order, Committed, Available) & CKEditor notes.
4. Part Superseding	Edit legacy part at `/parts/<pk>/edit/`, pick replacement.	Displays yellow warning badge on legacy part and "Supersedes" link on new part.